Air Quality Index (AQI) Prediction using Machine Learning in PySpark

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PYSPARK · ML

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Introduction: Why AQI Prediction Matters

The Problem

Air pollution is a leading global health risk, causing millions of premature deaths annually. AQI quantifies pollutant levels into a single actionable score.

The Opportunity

Predicting AQI enables proactive public health alerts, policy decisions, and early warnings for vulnerable populations.

The Approach

Machine learning on distributed data (PySpark) offers scalable, accurate AQI forecasting from multi-pollutant sensor data.

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Project Objectives

1

Data Pipeline

Build an end-to-end PySpark pipeline for ingesting and preprocessing large-scale air quality datasets.

2

Feature Engineering

Derive meaningful features from raw pollutant readings (PM2.5, PM10, NO_2, CO, O_3) to improve model accuracy.

3

Predictive Modeling

Train and evaluate a Linear Regression model in PySpark MLlib to predict AQI with high precision.

4

Interpretability

Analyze feature importance and model residuals to generate actionable environmental insights.

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DATASET

Dataset Description

- **Source:** UCI Machine Learning Repository — Beijing Multi-Site Air Quality Dataset
- **Duration:** Hourly readings across 12 monitoring stations (2013–2017)
- **Target Variable:** AQI (computed from pollutant concentrations)

Input Features

PM2.5
Fine particulates

PM10
Coarse particulates

NO_2
Nitrogen dioxide

CO
Carbon monoxide

O_3
Ground-level ozone

So2
Sulfur dioxide

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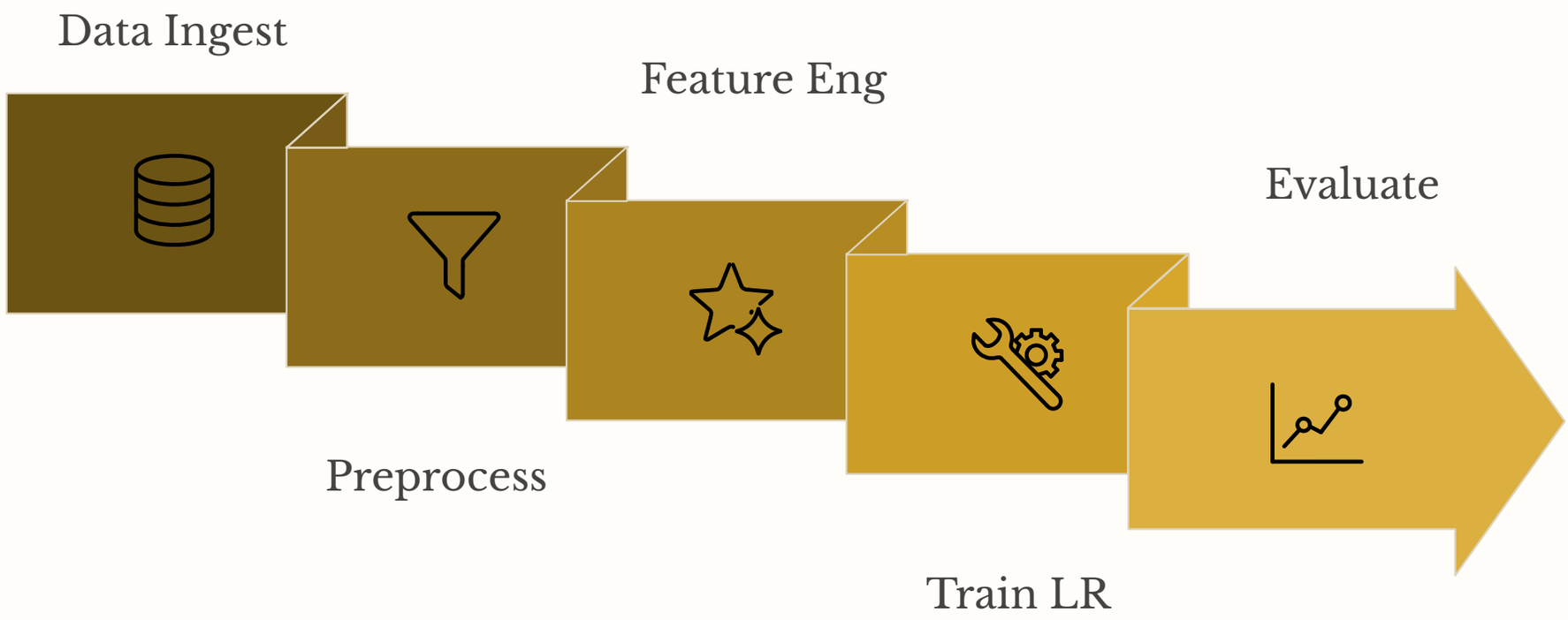
Methodology Overview



The end-to-end workflow leverages PySpark's distributed computing to handle large environmental datasets efficiently, from raw ingestion through model evaluation.

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Results: Model Performance

2.38

RMSE

Root Mean Squared Error on test set — low error indicates strong predictive accuracy.

0.978

R^2 Score

97.8% of AQI variance explained by the model — exceptional fit.

12

Stations

Monitoring sites across Beijing included in the dataset.

5yr

Data Span

Five years of hourly environmental readings used for training.

Key Observation: PM2.5 and NO_2 emerged as the strongest predictors of AQI, consistent with their dominant contribution to the AQI formula.

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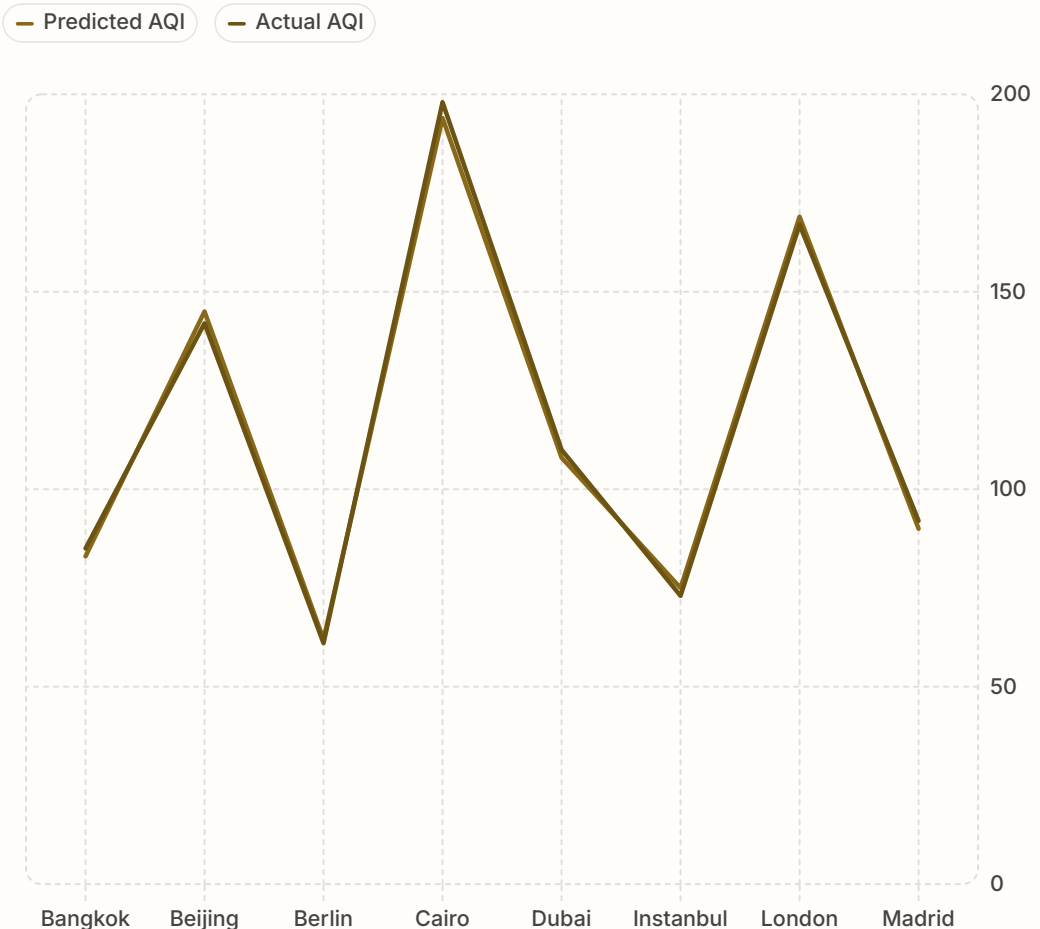
Visualization: Actual vs. Predicted AQI



Interpretation

Predicted values closely track actual AQI across the full range — from moderate (60–90) to unhealthy (150–200) levels.

City-wise AQI Comparison

AQI varied significantly across Beijing's 12 stations. Urban-center stations recorded consistently higher AQI, while suburban stations showed lower pollution levels — validating the model's spatial sensitivity.

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Conclusion

What We Achieved

A PySpark Linear Regression model achieved **R^2 = 0.978** and **RMSE = 2.38**, demonstrating that distributed ML can accurately predict AQI from multi-pollutant data at scale.

References

- UCI Machine Learning Repository — Beijing Air Quality Dataset
- Apache Spark MLlib Documentation (v3.4)
- WHO Air Quality Guidelines, 2021

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